

Characterizing the γ -ray sky using differentiable probabilistic programming

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Abstract

We use differentiable probabilistic programming, enabled by recent advances in machine learning, in order to account for the large model-space inherent to γ -ray analyses in a principled manner. Focusing on the ongoing mystery of the Galactic Center Excess (GCE), we construct a differentiable forward model and likelihood that liberally uses GPU acceleration and vectorization in order to simultaneously account for a range of possible spatial morphologies consistent with the Excess emission. Our setup allows for efficient inference over the large model space using variational methods. The goal of this work is to enable analyses that are more robust to specific modeling choices compared to traditional techniques, and to provide a framework for future analyses.

1. Introduction

Analysis of γ -ray data towards the Galactic Center involves a subtle interplay between observations and theoretical models. The complexity of cosmic-ray transport processes in the region, combined with the fact that we only have access to a two-dimensional projection of this inherently three-dimensional process, means that we typically cannot make precise predictions about crucial emission mechanisms. Consequently, scientific conclusions drawn from γ -ray analyses can depend strongly on specific assumptions made about the modeling of the signal and background, as well as manual choices involved in the data analysis pipeline.

The γ -ray Galactic Center Excess (GCE), first identified over a decade ago (Goodenough & Hooper, 2009) using public data from the *Fermi* -LATE telescope (Atwood et al., 2009), is an excess of photons in the Inner Galaxy whose physical origin is currently unknown. Although the GCE has properties broadly compatible with those expected from an-

ihilating dark matter (DM) (Goodenough & Hooper, 2009; Hooper & Goodenough, 2011; Daylan et al., 2016; Calore et al., 2015; Ajello et al., 2016), competing explanations in terms of a population of unresolved astrophysical point sources (PSs), in particular millisecond pulsars (MSPs), remain viable (Abazajian et al., 2014; Abazajian, 2011). Significant effort has gone into characterizing the GCE and understanding its origin, including through inferring the properties of unresolved PSs in the Inner Galaxy region. A particularly successful method in this direction, Non-Poissonian Template Fitting (NPTF), aims to characterize different populations of unresolved PSs using their photon statistics, via a likelihood-based analysis (Lee et al., 2015; 2016). In the context of the GCE, initial applications of this method showed overwhelming support for the hypothesis of the GCE being sourced by an unresolved population of PSs spatially correlated with the observed GCE morphology (Lee et al., 2016). However, more recent studies showed the method to be susceptible to systematic biases (Leane & Slatyer, 2019; 2020a;b; Chang et al., 2020) partly stemming from degeneracies between different emission components in the region (e.g., those spatially correlated with the GCE, centrally-concentrated stellar populations, and the Galactic disk), with results depending strongly on specific assumptions made about the spatial morphologies of components. *All NPTF analyses so far have assumed inflexible, spatially rigid templates, preventing a rigorous expression and exploration of this degeneracy.*

In this work, we use differentiable probabilistic programming to alleviate this shortcoming of traditional likelihood-based fitting techniques for analyzing γ -ray data. Using GPU-native acceleration and vectorization, we construct a forward model for γ -ray emission in the Galactic Center region that can generate high-resolution spatial templates for various emission components, including those associated with the Galactic Center excess signal, on the fly. This allows for analyses that are robust to specific analysis choices, and opens up the possibility of performing model-independent analyses using the framework. In the broader context of astrophysics, this work aims to motivate the use of differentiable probabilistic programming as a way to enable high-dimensional Bayesian inference for complex descriptions of diverse astrophysical data.

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2. Methodology

We describe our forward model and likelihood, its implementation as a differentiable probabilistic program, as well as the inference strategies used to perform Bayesian posterior inference over the model parameters characterizing various emission mechanisms for Galactic Center γ -ray data.

2.1. The likelihood and forward model

In traditional template fitting, photon counts in different pixels p are modeled as Poisson random variables, with mean given by a linear combination of templates T_i^p , indexed i ; $x^p \sim \text{Pois}(x^p | \sum_i A_i T_i^p)$. A_i are the template normalizations and represent the parameters of interest. The full map-levels likelihood factorizes into a produce of pixel-wise likelihoods, $p(x | \{A_i\}) = \prod_p p(x_p | \{A_i\})$.

In the presence of populations of unresolved point sources, the likelihood can be augmented to take into account the probability that a given pixel contains a point source from a given population, and the probability of that point source emitting a certain mean number of contributing photons. Point source populations can be independently characterized by a spatial distribution, specified by a template $T_{\text{PS},i}^p$, and a source-count distribution dN/dS , which jointly describes the distribution of photon counts S from individual PSs, $p(S|\theta_{\text{PS}})$, and their mean abundance, $\bar{n}_{\text{PS}}$; $dN/dS = \bar{n}_{\text{PS}} p(S | \theta_{\text{PS}})$.

Unfortunately, the inclusion of point source populations renders the likelihood intractable, since computing it would involve marginalizing over positions, counts, and cardinality of point sources from each population. The NPTF method (Lee et al., 2015; 2016) circumvents this issue by instead computing the pixel-wise likelihood, and approximating the full likelihood as $p(x | \bar{n}_{\text{PS}}, \theta_{\text{PS}}) = \prod_p p(x^p | \bar{n}_{\text{PS}}, \theta_{\text{PS}})$. See Mishra-Sharma et al. (2017) for further details on the method. We implement a differentiable version of the NPTF likelihood in the Jax framework (Bradbury et al., 2018), making liberal use of vectorization and automatic batching to efficiently compute the likelihood for multiple point source populations.

2.2. Flexible specification of point source populations

Previous analyses based on the NPTF framework assumed rigidly specified spatial templates for the distribution of point source populations; this is because generating new morphologies adds to considerable overhead in the likelihood calculation, rendering a traditional Monte Carlo sampling-based analysis difficult. In particular, given a morphology for the PS template, $\rho(R, z)$ (assumed cylindrically symmetric), computing the template expectation in a give pixel p involves doing a line-of-sight integral in

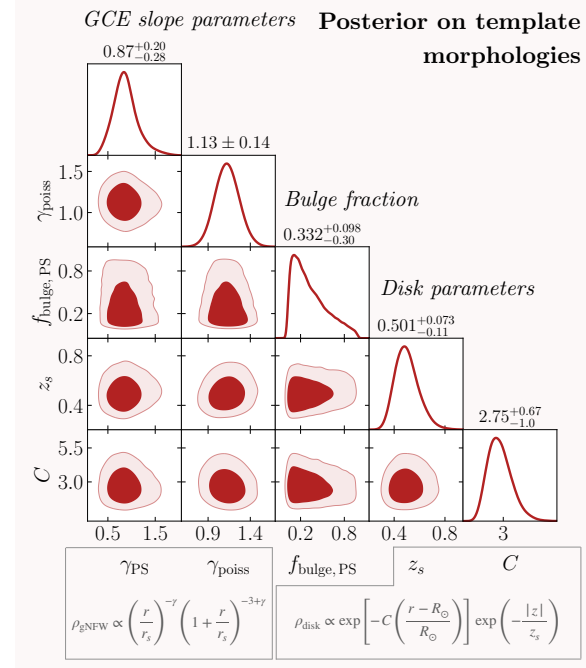


Figure 1. A figure.

the pixel direction, $T_{\text{PS}}^p = \int dl \rho(R(l, p), z(l, p))$. In Jax, the density over the line of sight as well across the pixel index can be efficiently vectorized, and the computation is just-in-time (JIT) compiled for efficient execution on a GPU.

This allows us to include parametric variations on different spatial templates that have been previously included in NPTF analyses, including templates for PSs correlated with the Galactic disk and the GCE, directly as part of the inference pipeline. We model a PS population with a generalized NFW profile, $\rho_{\text{gNFW}}(r) \propto (r/r_s)^{-\gamma} (1 + r/r_s)^{-3+\gamma}$ with the inner slope γ a free parameter to be inferred during the analysis. We also model a population of PSs spatially correlated with the Galactic disk, modeled as a double exponential, $\rho_{\text{disk}}(R, z) \propto (R/R_\odot)^B \exp[-C(R - R_\odot)/R_\odot] \exp(-|z|/z_s)$ with the parameter B , C , and the scale height z_s characterizing the disk morphology and to be inferred during the analysis.

2.3. Inference techniques

Since the entire pipeline is end-to-end differentiable, we can evaluate gradients of the log-likelihood function with respect to the parameters of interest θ characterizing the smooth as well as PS components, $\nabla_\theta \log \mathcal{L}(\theta)$. This opens up the possibility of using highly efficient gradient-based posterior inference techniques like Stochastic Variational Inference (SVI) (Hoffman et al., 2012) and Hamiltonian Monte Carlo (HMC) (Hoffman & Gelman, 2011). Unlike

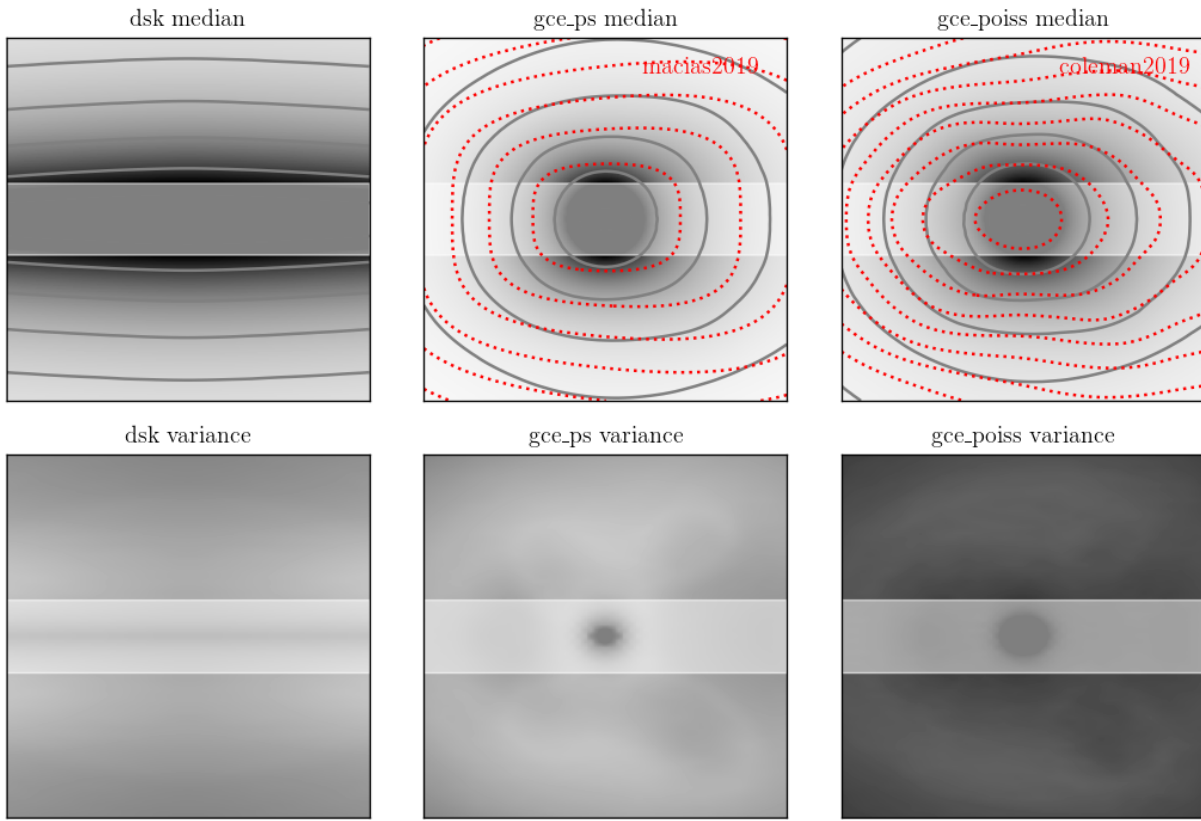


Figure 2. A figure.

traditional sampling techniques employed in the context of *Fermi* GCE analyses so far, these methods can easily scale up to high-dimensional parameter spaces. In particular, variational methods turn posterior inference into an optimization problem by fitting for a flexible functional ansatz on the posterior density and can easily deal with hundreds of parameters. This is in contrast to current public implementations of NPTF, which cannot efficiently include real-time generated templates at inference and are restricted to a model $\lesssim 30$ parameters with traditional Monte Carlo sampling techniques. We will wrap the constructed likelihood using the probabilistic programming framework NumPyro (Phan et al., 2019), which allows for flexible specification of complex models as well as performing inference on them using gradient-based inference techniques. Illustrative results for the posterior on the GCE and disk template parameters using a preliminary version of the NPTF likelihood written in Jax are shown in The scalability of our implementation will also support inclusion of a wider range of templates for Inner Galaxy PS populations and Galactic diffuse emission, although the development of improved diffuse models is beyond the scope of this proposal.

SM: Brief description of SVI.

3. Experiments and results

SM: Results of the experiment we will show (posteriors and such).

4. Discussion and future work

SM: A summary of our contributions.

- We can easily include multiple signal and background template descriptions in a principled manner.
- Scalability to higher number of parameters.
- Departure from rigid parameters – generate templates on-the-fly during inference.
- Much faster analysis.

Modern probabilistic programming languages like NumPyro treat distributions and random variables as first-class objects, allowing for the specification of models in a manner that is syntactically close to the PGM. The model can then be directly used for performing inference (by conditioning it on observed data) as well as producing simulations (via sampling the model in the forward direction). The latter will be used to effectively validate the analysis results via posterior-predictive checks, which assess the compatibility between simulated realizations from the data-conditioned model and real data. Strategies such as *enumeration* (Obermeyer et al.,

2019), which parallelize the likelihood computation over all possible combinations of the number of individual PSs, will be used to perform explicit marginalization while retaining the benefits of gradient-based posterior inference. This method will have several advantages over all existing approaches. Unlike the NPTF method, it will be able to fully account for pixel-to-pixel correlations. In contrast to machine learning-based approaches, being likelihood-based it will not require the use of black-box feature extractors, and additionally will easily scale to the larger model-spaces described in previous stages of the project since it will not rely on the use of pre-specified training data that is required to have support over the entire model-space.

Directly building on the work done so far, our next goal will be to characterize the γ -ray PS populations in the Galactic Center in a fully signal model-independent manner, by scanning over a space of spatial *functions* on the sky map, as opposed to using parametric descriptions of signal templates. We will use Gaussian processes to place a prior over this function space, which will allow us to specify a scale on which we expect the morphology of the signal to vary, leaving the per-pixel normalizations otherwise free. The posteriors over both the smooth as well as unresolved PS population templates will provide an important cross check with the fixed-template descriptions used so far, and will, for the first time, deliver a model-independent characterization of the spatial population of unresolved PSs in the Inner Galaxy. A similar Gaussian process-based technique was successfully implemented in Mishra-Sharma & Cranmer (2020) to augment the morphology of the diffuse foreground template. The inclusion of flexible spatial templates in the pipeline is enabled by the highly efficient and differentiable version of the NPTF likelihood.

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